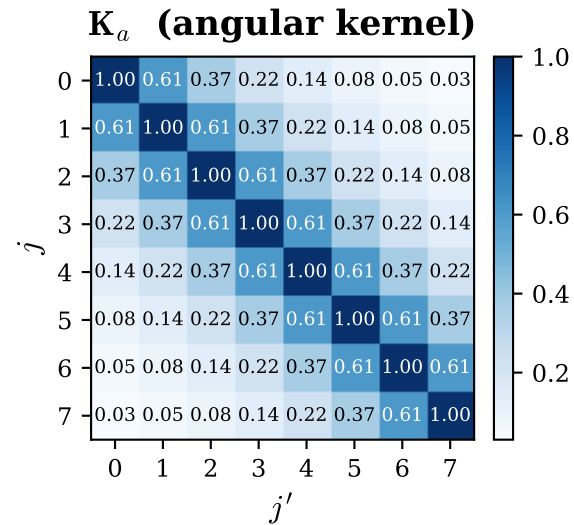
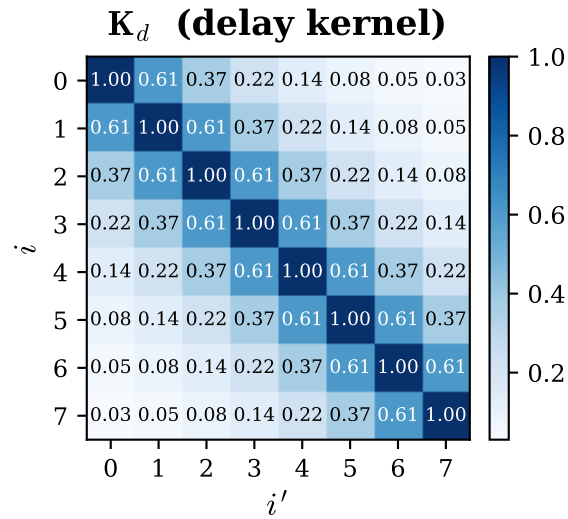


Compression Difficulty Metric ζ : Compact vs Diffuse CSI ($N=8, \tau=2.0$)

Locality
Kernels



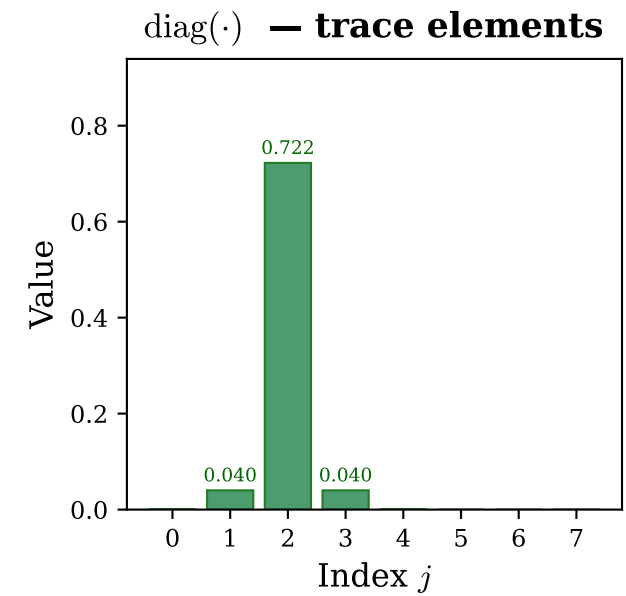
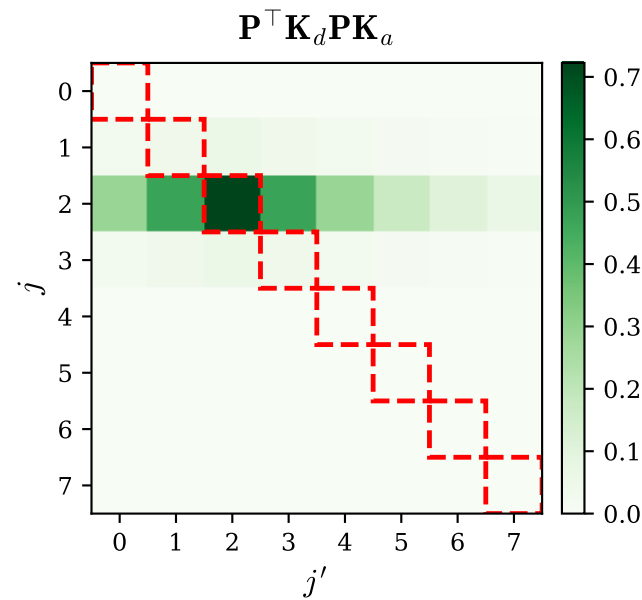
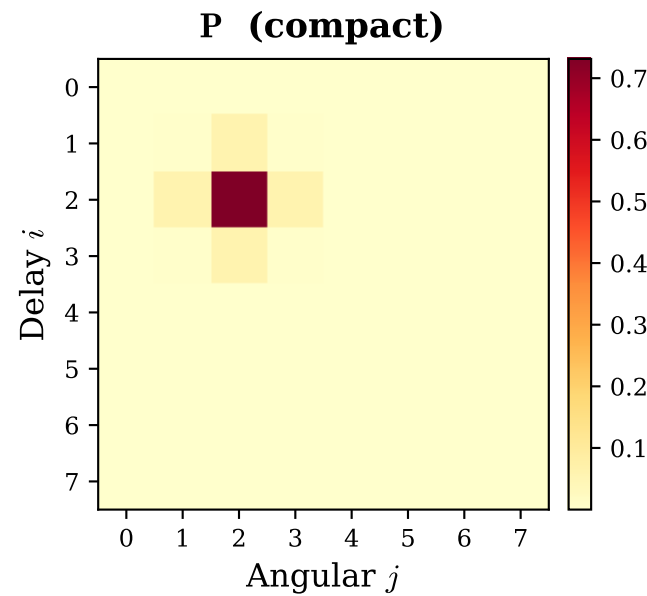
$$\zeta(\mathbf{X}_a) = 1 - \text{tr}(\mathbf{P}^\top \mathbf{K}_d \mathbf{P} \mathbf{K}_a)$$

$$\mathbf{P}_{ij} = \frac{|X_{ij}|^2}{\sum_{u,v} |X_{uv}|^2} \quad (\text{normalized energy})$$

$$[\mathbf{K}]_{i,i'} = e^{-|i-i'|/\tau} \quad (\text{locality kernel})$$

Small $\zeta \rightarrow$ compact, easy to compress
Large $\zeta \rightarrow$ diffuse, hard to compress

Compact
CSI

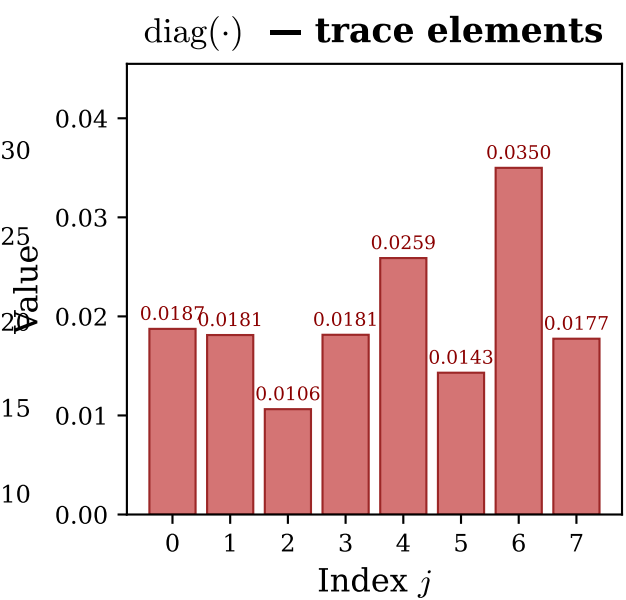
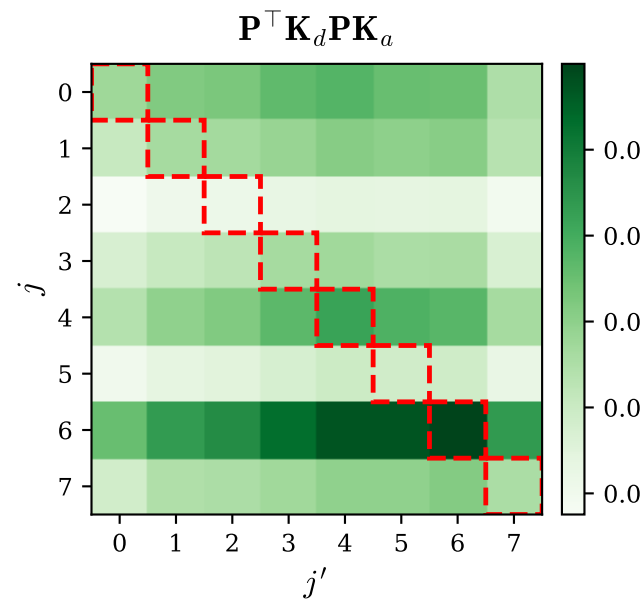
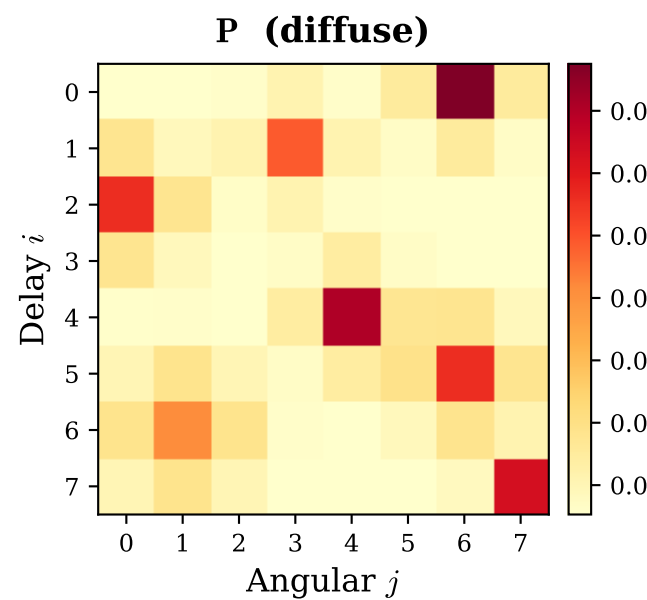


$$\text{tr} = \mathbf{0.8025}$$

$$\zeta = \mathbf{0.1975}$$

$\Leftarrow$ Easy to compress

Diffuse
CSI



$$\text{tr} = \mathbf{0.1586}$$

$$\zeta = \mathbf{0.8414}$$

$\Leftarrow$ Hard to compress